Integration and Kinematics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

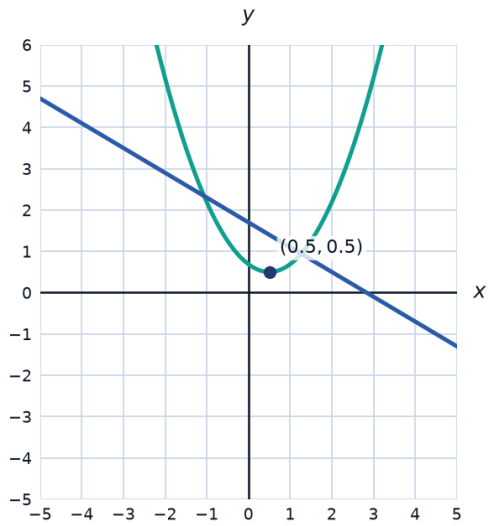
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Integration & Kinematics Practice Exercises

A cuboid has a length of USD 8$cm, a width of USD 5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

2 [Foundational] • Target: Integration & Kinematics Practice Exercises

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.



3 [Foundational] • Target: Integration & Kinematics Practice Exercises

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

4 [Foundational] • Target: Integration & Kinematics Practice Exercises

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

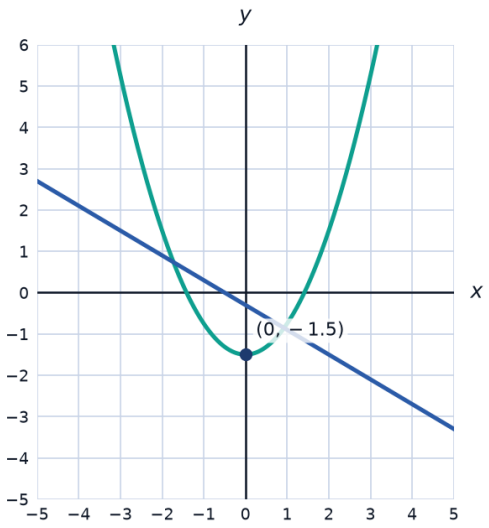
5 [Foundational] • Target: Integration & Kinematics Practice Exercises

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

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6 [Foundational] • Target: Integration & Kinematics Practice Exercises

A car travels a distance of $d = 180$ km correct to the nearest 10 km in a time of $t = 2.4$ hours correct to 1 decimal place. Calculate the upper bound for the average speed of the car in km/h.



7 [Foundational] • Target: Integration & Kinematics Practice Exercises

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration & Kinematics Practice Exercises

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

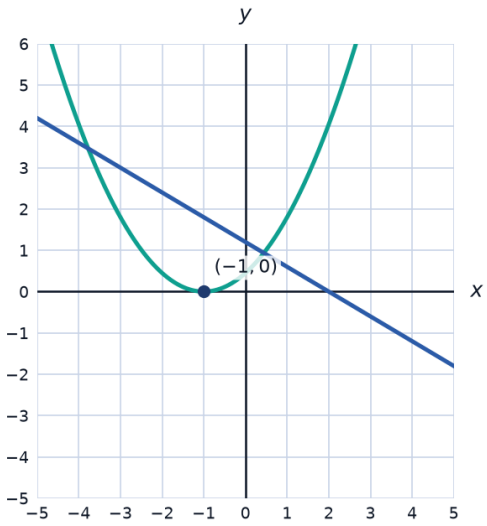
9 [Intermediate] • Target: Integration & Kinematics Practice Exercises

The table shows the time, t minutes, taken by 50 students to complete a homework task. Intervals: $0 < t \le 10$ (freq: 8), $10 < t \le 20$ (freq: 14), $20 < t \le 30$ (freq: 18), $30 < t \le 40$ (freq: 10). Calculate an estimate of the mean time taken.

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10 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.



11 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of USD 2000 to exceed USD 3000?

12 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A student computes the volume of a cylinder using $r = 3.2\ \text{cm}$ and $h = 7.85\ \text{cm}$. The raw calculator display shows $252.518...$. How should the final answer be reported according to Cambridge AS Level guidelines?

13 [Intermediate] • Target: Integration & Kinematics Practice Exercises

In a survey of 60 students, 35 study French, 28 study German, and 12 study both languages. Find (a) the number of students who study neither language, and (b) the probability that a student chosen at random studies French given that they study German.

14 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

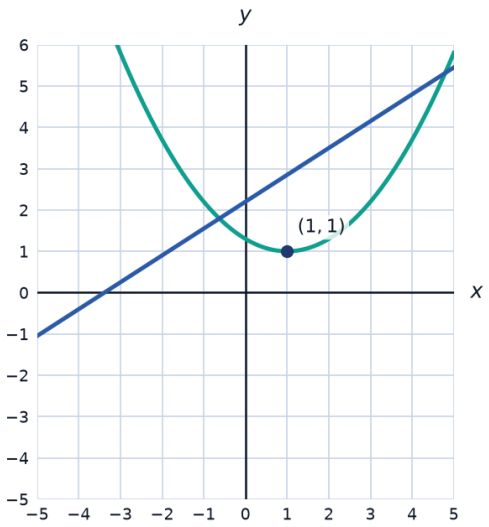
15 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

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16 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In a cyclic quadrilateral $ABCD$, angle $ABC = (2x + 15)^\circ$ and angle $ADC = (3x - 10)^\circ$. Find the value of x and hence the size of angle ABC. Give a geometric reason for your equation.



17 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.

18 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

19 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A discrete random variable Y has $E(Y) = 5$ and $\text{Var}(Y) = 4$. Calculate $E(Y^2)$ and $\text{Var}(10 - 3Y)$.

20 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

A continuous dataset is grouped into class intervals. The interval $20 \le x < 35$ has a frequency of 45. Find the frequency density for this class.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Integration & Kinematics Practice Exercises

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$. The velocity of the particle is given by $v = \frac{ds}{dt}$. Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

a. t=1 s, s=6 m

b. t=2 s, s=4 m

c. t=3 s, s=2 m

d. t=1 s, s=2 m

22 [Foundational] • Target: Integration & Kinematics Practice Exercises

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$

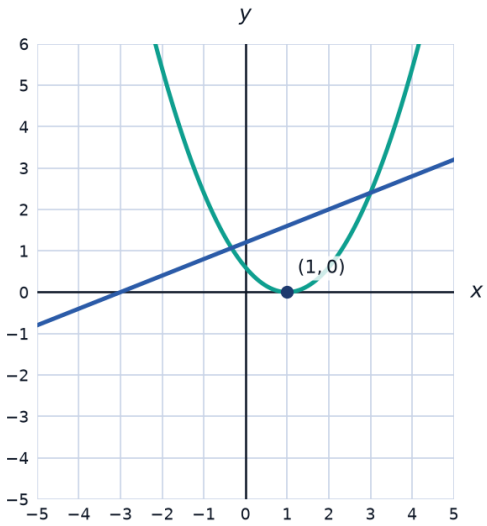
b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

23 [Foundational] • Target: Integration & Kinematics Practice Exercises

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.



a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

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24 [Foundational] • Target: Integration & Kinematics Practice Exercises

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

25 [Foundational] • Target: Integration & Kinematics Practice Exercises

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

a. A = 1/6, D = 5/3

b. A = 1/6, D = 10/3

c. A = 1/3, D = 5/3

d. A = 1/3, D = 10/3

26 [Foundational] • Target: Integration & Kinematics Practice Exercises

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

a. 16 m

b. 20 m

c. 24 m

d. 28 m

27 [Foundational] • Target: Integration & Kinematics Practice Exercises

Find the exact value of the definite integral $\int_0^{\ln 2} e^{2x}\,dx$.

a. $\frac{3}{2}$

b. 3

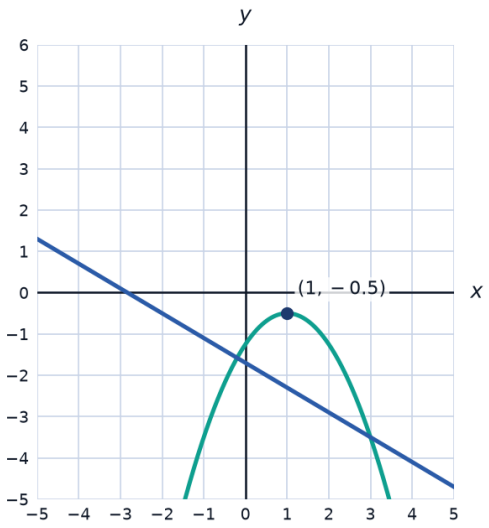
c. $\frac{1}{2}$

d. $\frac{7}{2}$

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28 [Foundational] • Target: Integration & Kinematics Practice Exercises

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.



a. -6

b. 6

c. 3

d. -3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

30 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. 4/3

c. -3/4

d. -4/3

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31 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. 5

b. 7

c. 12

d. 25

32 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■3, 4■

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

33 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Using the trapezium rule with 3 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 1, y_1 = 1, y_2 = 2, y_3 = 10$.

a. $\frac{17}{2}$

b. 7

c. $\frac{103}{12}$

d. $\frac{19}{2}$

34 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = t^2 - 4t$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s (excluding $t = 0$)

b. $t = 5$ s (excluding $t = 0$)

c. $t = 0$ s only

d. $t = 4$ s (excluding $t = 0$)

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35 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A particle has initial velocity $u = 10$ m/s and constant acceleration $a = 4$ m/s^2. Find its velocity after $t = 3$ s.

a. 22

b. 21

c. 23

d. 14

36 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Find $\int x(x^2 + 1)^2\,dx$ using substitution.

a. $\frac{x^2(x^2+1)^2}{2} + C$

b. $\frac{x^6}{6} + \frac{x^4}{2} + \frac{x^2}{2} + C$

c. $\frac{2(x^2+1)^3}{3} + C$

d. $\frac{(x^2+1)^3}{3} + C$

37 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A particle has initial velocity $u = 3$ m/s and constant acceleration $a = 3$ m/s^2. Find its speed after travelling a displacement $s = 11$ m.

a. $\sqrt{79}$

b. $\sqrt{71}$

c. 75

d. $\sqrt{75}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A particle has initial velocity $u = 4$ m/s and constant acceleration $a = 4$ m/s^2. Find its displacement after $t = 6$ s.

a. 24

b. 98

c. 94

d. 96

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39 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A particle moves so that its velocity at time t s is $v = 8t + 3$ m/s. Find the acceleration at $t = 0$ s.

a. 13

b. 10

c. 8

d. 6

40 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s in $t = 6$ s. Find its acceleration.

a. 1

b. 3

c. 2

d. 12/7

41 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

a. -1

b. 1

c. $1 + e^2$

d. 2

42 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A particle moves so that its velocity at time t s is $v = 2t$ m/s. Find an expression for its displacement s in terms of t.

a. $3t^2 + C$

b. $t^2 + C$

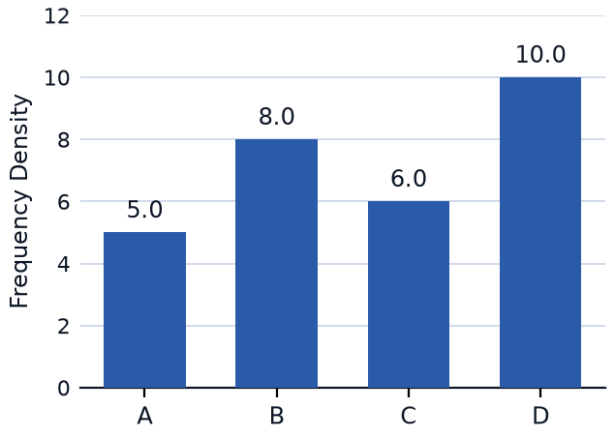
c. 1+C

d. $2t^2 + C$

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43 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

On a histogram, a class interval is $20 \leq x < 30$ with a frequency of 25. What is the frequency density for this bar?



a. 2.50

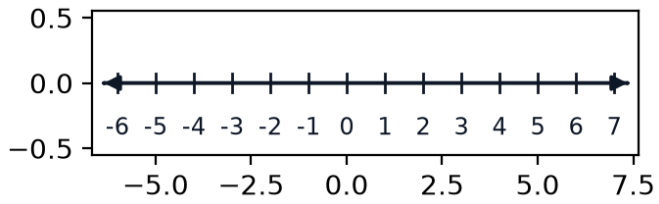
b. 5.00

c. 25

d. 10

44 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The diagram shows a number line with points marked. What is the value of x?



a. -5

b. -3

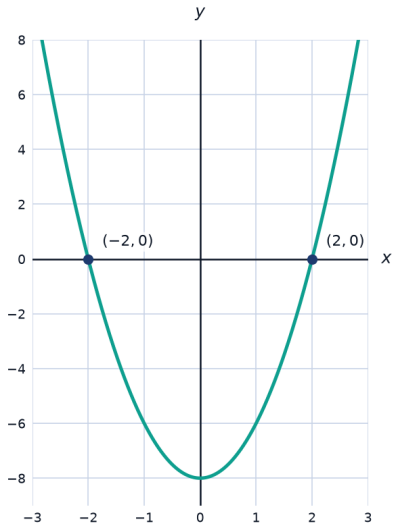
c. 1

d. 5

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45 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The equation $y = 2x^2 - 8$ crosses the x-axis at:



a. $x = \pm 2$

b. $x = \pm 4$

c. $x = 2$

d. $x = \pm\sqrt{8}$

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